

# SLIP – 7

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## Section I: Data Structures [15 Marks]

**Q 1) Accept n sorted values in array from user. Accept a value x from user and use binary search algorithm to check whether the number is present in sorted array or not and output the position if the number is present.**

**Answer:**

```
#include <stdio.h>

int main()
{
    int a[100], n, i, x, low, high, mid, pos = -1;

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
        return 0;
    }

    printf("Enter %d sorted values:\n", n);
    for(i = 0; i < n; i++)        scanf("%d",
    &a[i]);

    printf("Enter value x to search: ");
    scanf("%d", &x);

    low = 0;
    high = n - 1;

    while(low <= high)
    {
        mid = (low + high)
/ 2;

        if(a[mid] == x)
        {
            pos = mid + 1;
            break;
        }
        else
        if(a[mid] < x)
            low = mid + 1;
        else
            high = mid - 1;
    }
}
```

```

        if(pos != -1)           printf("Number found at
position %d", pos);           else           printf("Number
not found");

        return 0;
}

```

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## Section II: Database Management Systems [15 Marks]

**Q 2) Consider the following database( primary keys are underlined).**

**Item (itemno, name, quantity)**

**Supplier (sno, name, city)**

**The relationship is as follows: Item-Supplier: many-to-many with rate as a descriptive attribute.**

**Q 2(A)**

**A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]**

**Answer:**

```

CREATE TABLE Item(      itemno
INT PRIMARY KEY,      name
VARCHAR(50),      quantity INT
);

CREATE TABLE Supplier(
sno INT PRIMARY KEY,
name VARCHAR(50),      city
VARCHAR(50)
);

CREATE TABLE Item_Supplier(      itemno
INT REFERENCES Item(itemno),      sno INT
REFERENCES Supplier(sno),      rate
NUMERIC(10,2),
      PRIMARY KEY(itemno, sno)
);

INSERT INTO Item VALUES
(1, 'Mouse', 50),
(2, 'Keyboard', 8),

```

```
(3, 'Monitor', 20);
```

```
INSERT INTO Supplier VALUES  
(1, 'Mahesh', 'Mumbai'),  
(2, 'Suresh', 'Pune'), (3, 'Mohan', 'Nashik');
```

```
INSERT INTO Item_Supplier VALUES  
(1, 1, 300),  
(2, 2, 450),  
(3, 3, 5000),  
(1, 2, 280);
```

## Q 2(B)

### B. Execute the following queries in PostgreSQL (any 3) [6 marks]

i. Change the quantity of item 'Mouse' to 80.

#### Answer:

```
UPDATE Item  
SET quantity = 80  
WHERE name = 'Mouse';
```

ii. List the details of the suppliers whose name begins with the alphabet 'M'.

#### Answer:

```
SELECT * FROM Supplier WHERE  
name LIKE 'M%';
```

iii. Display the total count of items.

#### Answer:

```
SELECT COUNT(*) FROM Item;
```

iv. List the names of suppliers who do not live in Pune city.

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## Q 2(C)

### C. Write a query to display the name of supplier who supplies Item with minimum rate. [4 marks]

#### Answer:

```
SELECT S.name
```

```
FROM Supplier S
JOIN Item_Supplier ISU ON S.sno = ISU.sno
WHERE ISU.rate = (SELECT MIN(rate) FROM Item_Supplier);
```

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**OR**

**C. Create a view to display supplier-wise list of items.**

**Answer:**

```
CREATE VIEW Supplier_Wise_Items AS
SELECT S.name AS supplier_name,
       I.name AS item_name
FROM Supplier S
JOIN Item_Supplier ISU ON S.sno = ISU.sno
JOIN Item I ON I.itemno = ISU.itemno
ORDER BY S.name;
```

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**Q 3) Viva [5 marks]**